

Jimmy Ding

jimmydingjk@gmail.com | +XX XX-XXX XXXX | Malaysia | jimmyding.dev | [LinkedIn](#) | [GitHub](#)

Experience

AI Systems Research and Development Intern (Full-Time) | [OpenSys \(M\) Berhad](#) **Oct 2024 – Jan 2025**

- Prototyped **AI components** for internal workflow automation, comparing commercial APIs with open-source alternatives across reliability, output quality, and deployment feasibility.
- Designed **structured test cases and error-analysis reports** to identify model failure modes in production-like scenarios.
- Integrated selected AI components into existing **internal workflows**, supporting testing, validation, and deployment handoff.

Selected Projects

Offline Signature Verification | [GitHub](#) | [Write-Up](#) | [Demo](#) **Oct 2025 – Jun 2026**

- Built a **PyTorch** deep metric learning pipeline for **writer-independent** offline signature verification on CEDAR, achieving **0.9284 ROC-AUC** on skilled forgery detection using transfer learning with **EfficientNetV2** and **signer-aware batch sampling**.
- Designed and evaluated L_{SC+} , a custom metric-learning loss that improves separation between genuine and forged signatures while preserving intra-class signature variation.
- Deployed the model in a containerised, stateless real-time inference demo, self-hosted via **Docker** and **Tailscale VPN**, for live signature verification.

mediadl – Cross-Platform Media Archiving CLI | [GitHub](#) | [Releases](#) **Jun 2026**

- Built a Rust CLI supporting **3 media workflows** (audio, video, and CSV batch downloads) with structured creator and collection-based output through yt-dlp and FFmpeg.
- Designed a modular **2-crate Cargo workspace** with **10 configurable options** for media formats, quality, output templates, thumbnails, retries, and parallel downloads.
- Automated GitHub Actions builds and release packaging for **4 platform targets** across Linux, Windows, and macOS.

Education

Bachelor of Computer Science (Honours) | [Universiti Tunku Abdul Rahman](#) | Malaysia **May 2023 – Jun 2026**

- **CGPA: 3.96/4.00**
- **Scholarships & Awards:** Top Achiever 100% Scholarship | China Ambassador Scholarship | President's List Awards.

Skills

- **AI/ML:** PyTorch | OpenCV | scikit-learn | NumPy | pandas | TensorBoard | Streamlit | Cursor | Claude Code
- **Programming Languages:** Python | Rust | C++ | Java | Lua | SQL | JavaScript | TypeScript | HTML | CSS
- **Databases:** Microsoft Access | MySQL | PostgreSQL | MongoDB
- **Cloud, Infrastructure, DevOps:** AWS | Alibaba Cloud | Git | CI/CD | Docker | Nginx | Linux

Hackathons & Competitions

Applied LLM Engineering | [Cursor X Anthropic Hackathon](#) | [Monash University](#) | Malaysia **Dec 2025**

- Built an **LLM-powered** budget journal with natural-language journal entries to generate **personalised financial insights**.

Cryptocurrency Prediction Model | [UMHackathon](#) | [University of Malaya](#) | Malaysia **Apr 2025**

- Built a forecasting pipeline combining **LSTM-based price prediction**, **VADER news sentiment analysis**, and **Hidden Markov Models** for market-regime detection.

Fraud Detection and Prevention Plugin | [Hack Attack](#) | [Universiti Tunku Abdul Rahman](#) | Malaysia **Aug 2024**

- Built a browser plugin leveraging **web scraping**, **a custom sentiment classification model**, and **Alibaba Cloud database services** to aggregate and summarise product reviews in real-time.

Open Source

Quartz5 Plugins | [quartz5-drawer](#) | [quartz5-clickable-images](#) **Dec 2025**

- Developed and published two TypeScript plugins for the Quartz5 static site generator; an original mobile table of contents drawer and an image lightbox zoom ported from Quartz4, each with GitHub Actions CI/CD.